05-07-2026

Agenda: ML-15

- Naive Bayes Algo (2 methods)

Naive Bayes Algo:

A probabilistic machine learning algo. It starts with a baseline assumption/guess & then updates its belief every times it sees a new piece of information (features), multiplying the guess together to reach the final verdict.

Naive Bayes

← assumes every feature is completely independent of the other

→ Bayes Theorem

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

$P(X|Y) \rightarrow P(\text{humidity} / \text{Rain})$

$P(Y) \rightarrow P(\text{Rain})$

$P(X) \rightarrow P(\text{humidity})$

$P(Y|X) \rightarrow$ posterior

$P(\text{Rain} / \text{humidity})$

$Y \rightarrow$ weather (Rain)

$X \rightarrow$ humidity

$X_1 \rightarrow$ wind speed

$X_2 \rightarrow$ Cloud

Base assumption

$$\text{Total\_score} = P(\text{Rain}) \times P(\text{humidity} / \text{Rain}) \times \boxed{P(\text{windspeed} / \text{Rain})} \times P(\text{cloud} / \text{Rain})$$

why Naive Bayes?

- → Data is categorical
- → we want probabilistic reasoning
- → we want something that learns from frequency count
- → we want fast & interpretable solution.

NB → classification → probabilistic model
→ supervised ML Algo

Dataset : (Tennis)

outlook	Temp	play
Sunny	HOT	NO
Sunny	Mild	NO
Cloudy	HOT	Yes
Rainy	Mild	Yes
Rainy	Cool	Yes
cloudy	cool	Yes

prob of Rainy & HOT ?

Baseline Assuption!

$$P(\text{yes}) = \frac{4}{6} = 0.66$$

$$P(\text{NO}) = \frac{2}{6} = 0.33$$

$P(\text{yes}) \to 0.66$

outlook	outlook_calc	Temp	Temp_calculation
Cloudy	2/4 → 0.5	Hot	1/4 → 0.25
Rainy	2/4 → 0.5	mild	1/4 → 0.25
Sunny	0/4 → 0	cool	2/4 → 0.5

$P(\text{NO}) \to 0.33$

outlook	outlook_calc	Temp	Temp_calculation
Cloudy	0/2 → 0	Hot	1/2 → 0.5
Rainy	0/2 → 0	mild	1/2 → 0.5
Sunny	2/2 → 1	cool	0/2 → 0

$$P(\text{Yes} \mid \text{Rain}, \text{Hot}) \propto P(\text{Yes}) \times P(\text{Rain} \mid \text{Yes}) \times P(\text{HOT} \mid \text{Yes})$$

$$= 0.66 \times 0.5 \times 0.25$$

$$= 0.0825$$

$$P(\text{No} \mid \text{Rain}, \text{HOT}) \propto P(\text{No}) \times P(\text{Rain} \mid \text{No}) \times P(\text{HOT} \mid \text{No})$$

$$= 0.33 \times 0 \times 0.5$$

$$= 0$$

$$P(\text{Yes} \mid \text{Rain}, \text{HOT}) > P(\text{No} \mid \text{Rain}, \text{HOT}) = \underline{\underline{\text{Yes}}}$$

→ prob(play) | Sunny & cool

$$P(\text{Yes} \mid \text{Sunny}, \text{cool}) \propto P(\text{Yes}) \times P(\text{Sunny} \mid \text{Yes}) \times P(\text{cool} \mid \text{Yes})$$

$$= 0.66 \times 0 \times 0.5$$

$$= 0$$

$$P(\text{No} \mid \text{Sunny}, \text{cool}) \propto P(\text{No}) \times P(\text{Sunny} \mid \text{No}) \times P(\text{cool} \mid \text{No})$$

$$= 0.33 \times 1 \times 0$$

$$= 0$$

$P(\text{Yes} \mid \text{Sunny}, \text{cool})$ ↘ $P(\text{No} \mid \text{Sunny}, \text{cool})$ ↙

Both are 0 → zero frequency prob

To solve above, we use something called as **smoothing**

$$\frac{\text{numerator} + 1}{\text{denominator} + (\text{no. of categories})}$$

→ we add 1 to every count → numerator

→ divide by # samples + # categories

to keep prob balance

100,000

outlook	outlook_calc	outlook_new	Temp	Temp_calculation	temp_new
cloudy	2/4 → 0.5	2+1/4+3 → 0.429	Hot	1/4 → 0.25	1+1/4+3 → 0.286
Rainy	2/4 → 0.5	2+1/4+3 → 0.429	mild	1/4 → 0.25	1+1/4+3 → 0.286
Sunny	0/4 → 0	0+1/4+3 → 0.143	cool	2/4 → 0.5	2+1/4+3 → 0.429
cloudy	0/2 → 0	0+1/2+3 → 0.2	Hot	1/2 → 0.5	1+1/2+3 → 0.4
Rainy	0/2 → 0	0+1/2+3 → 0.2	mild	1/2 → 0.5	1+1/2+3 → 0.4
Sunny	2/2 → 1	2+1/2+3 → 0.6	cool	0/2 → 0	0+1/2+3 → 0.2

p(Yes) → 4/6 → 0.66

p(NO) → 2/6 → 0.33

→ prob(play) | sunny & cool

$$p(\text{yes} \mid \text{sunny}, \text{cool}) \propto p(\text{yes}) \times p(\text{sunny} \mid \text{yes}) \times p(\text{cool} \mid \text{yes})$$

$$= 0.66 \times 0.143 \times 0.429$$

$$= 0.040$$

$$p(\text{No} \mid \text{sunny}, \text{cool}) \propto p(\text{No}) \times p(\text{sunny} \mid \text{No}) \times p(\text{cool} \mid \text{No})$$

$$= 0.33 \times 0.6 \times 0.2$$

$$= 0.0396$$

$p(\text{yes} \mid \text{sunny}, \text{cool})$ $\quad$ $p(\text{No} \mid \text{sunny}, \text{cool})$

Normalize to get the final prob :-

$$\text{sum} = 0.04 + 0.039 = 0.0796$$

$$p(\text{yes} \mid \text{sunny}, \text{cool}) = \frac{0.04}{0.0796} \approx 0.5025$$

$$p(\text{No} \mid \text{sunny}, \text{cool}) = \frac{0.039}{0.0798} \approx 0.4974$$

$$p(\text{yes} \mid \text{sunny}, \text{cool}) > p(\text{No} \mid \text{sunny}, \text{cool})$$

features → categorical
target → categorical
→ categorical Naive Bayes

Next topic :-

Dataset → numerical features → categorical NB?
→ Categorical Target Column →

→ Gaussian Naive Bayes
↓
assume
the continuous
number follows a
Gaussian (normal) distribution

height (cm)	Taller/Not
	Yes
	No

$$\text{prob} = \frac{1}{\infty}$$

step 1: learn the distribution for each class

mean(μ)   variance(σ^2)

step 2: Guassian PDF formula (finding the likelihood)

$$P(X|Y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) = N(x, \mu, \sigma^2)$$

likelihood of X   given Y

π → $\frac{22}{7}$ | 3.14

exp → e → 2.718

we compute :

$$P(Y|X) \propto P(Y) \prod N(x, \mu, \sigma^2)$$

product operator

for loop for multiplication

Dataset:

id	x_1	x_2	Y
A	1.0	2.0	Yes
B	2.0	1.0	Yes
C	1.5	1.8	Yes
D	3.0	3.0	NO
E	3.5	2.8	NO
F	2.9	3.2	NO

Step 1: mean, variance, std

$\mu(x_1 = Yes) = 1 + 2 + 1.5/3 \rightarrow 1.5 \leftarrow \mu_{Yes,1}$

$\mu(x_1 = NO) = 3 + 3.5 + 2.9/3 \rightarrow 3.133 \rightarrow \mu_{NO,1}$

$\mu(x_2 = Yes) = 2 + 1 + 1.8/3 \rightarrow 1.6 \rightarrow \mu_{Yes,2}$

$\mu(x_2 = NO) = 3 + 2.8 + 3.2/3 \rightarrow 3.0 \rightarrow \mu_{NO,2}$

$\sigma^2 = \Sigma (x - \mu)^2 / n \leftarrow$ variance

$(Yes)(x_1) = (1 - 1.5)^2, (2 - 1.5)^2, (1.5 - 1.5)^2$

$= (-0.5)^2, (0.5)^2, (0)^2$

sum $= 0.25 + 0.25 + 0 \rightarrow 0.5/3 \rightarrow 0.166$

$(NO)(x_1) =$ sum $\rightarrow 0.2066 \rightarrow 0.2066/3 \rightarrow 0.0688$

$(Yes)(x_2) =$ sum $\rightarrow 0.56 \rightarrow 0.56/3 \rightarrow 0.186$

$(NO)(x_2) =$ sum $\rightarrow 0.08 \rightarrow 0.08/3 \rightarrow 0.0266$

Step 2

input $\rightarrow$ $x = (\overset{x_1}{2.0}, \overset{x_2}{2.0})$

$\rightarrow$ calculate gaussian pdf for each feature $\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

$(Yes)(x_1 = 2.0) = \underbrace{\frac{1}{\sqrt{2\pi\, 0.166}}}_{\approx 0.9772} \cdot \exp\left(\underbrace{-\frac{(2 - 1.5)^2}{2 \times 0.166}}_{0.472}\right)$

$p(x_1 | \text{yes}) \approx 0.9722 \times 0.472 \approx 0.4616$

$p(x_2 | \text{yes}) \approx 0.6009$

$p(x_1 | \text{No}) \approx 1.36 \times 10^{-4}$

$p(x_2 | \text{No}) \approx 1.68 \times 10^{-8}$

Joint prob:

$x_1(Y) . x_2(Y) \rightarrow \text{yes}$ $\quad x_1(N) . x_2(N) \rightarrow \text{No}$

$p(x_1 | \text{No}) \times p(x_2 | \text{No}) \longrightarrow p(x | \text{No})$

$1.36 \times 10^{-4} \times 1.68 \times 10^{-8}$

$\approx 2.29 \times 10^{-12}$

$p(x_1 | \text{yes}) \times p(x_2 | \text{yes}) \rightarrow p(x | \text{yes})$

0.4616×0.6009

≈ 0.2774

step 3:

$p(\text{yes} | x) = p(y) \cdot p(x | \text{yes}) = 0.5 \times 0.2774 \approx 0.1387$

$p(\text{No} | x) = p(n) \cdot p(x | \text{No}) = 0.5 \times 2.29 \times 10^{-12} \approx 1.145 \times 10^{-12}$

step 4: normalise to get class prob

$\text{sum} = 0.1387 + 1.145 \times 10^{-12}$

$= 0.1387\ldots$

$$p(\text{Yes}|x) \approx \frac{0.1387}{0.1387\ldots} \approx 0.99999\ldots 1$$

$$p(\text{No}|x) \approx \frac{1.145 \times 10^{-12}}{0.1387\ldots} \approx 8.26 \times 10^{-12}$$

Decision : predict → Yes (very strongly)

10 features → $x_1 \ldots\ x_{10}$

$$p(\text{Yes}|x) = p(\text{Yes}) \times p(x_1|\text{Yes}) \times p(x_2|\text{Yes}) \times \ldots \times p(x_{10}|\text{Yes})$$

$$0.025 \times 0.3 \times 0.9 \times 0.1$$

↙ small decimal number

0.0000000000000000 . . .

↙ computer have a limit on how small a decimal they can store.

$$p(\text{Yes}|x) = \log\left(p(\text{Yes}) \times p(x_1|\text{Yes}) \times p(x_2|\text{Yes}) \times \ldots \times p(x_{10}|\text{Yes})\right)$$

$$\text{log-score} = \log\left(p(\text{Yes}) \times p(x_1|\text{Yes}) \times p(x_2|\text{Yes}) \times \ldots \times p(x_{10}|\text{Yes})\right)$$

$$\log(A \times B) = \log(A) + \log(B)$$

$$\text{log-score} = \log(p(\text{Yes})) + \log(p(x_1|\text{Yes})) + \log(p(x_2|\text{Yes})) + \ldots + \log(p(x_{10}|\text{Yes}))$$

→ numerical stability

problem statement:

features → category → categorical Naive Bayes
→ Numerical → Gaussian Naive Bayes

x_1 (categorical) (weather) x_2 (numerical) (temperature) y (Target) (play)

$$P(y \mid \text{sunny}, 72^\circ F)$$

$$P(\text{Yes} \mid \text{sunny}, 72^\circ F) \propto P(\text{Yes}) \times P(\text{sunny} \mid \text{Yes}) \times P(72^\circ F \mid \text{Yes})$$

$$P(\text{No} \mid \text{sunny}, 75^\circ F) \propto P(\text{No}) \times \underbrace{P(\text{sunny} \mid \text{No})}_{\text{categorical likelihood}} \times \underbrace{P(72^\circ F \mid \text{No})}_{\text{numerical likelihood}}$$

Assignment:

sklearn → categorical NB
→ Gaussian NB

Assignment : Think, how to solve with Naive bayes when features are both Categorical & Numerical using either Categorial/Gaussian NB

mixed-naive-bayes → MixedNB